

Association Rules mining

Princess of Data Mining

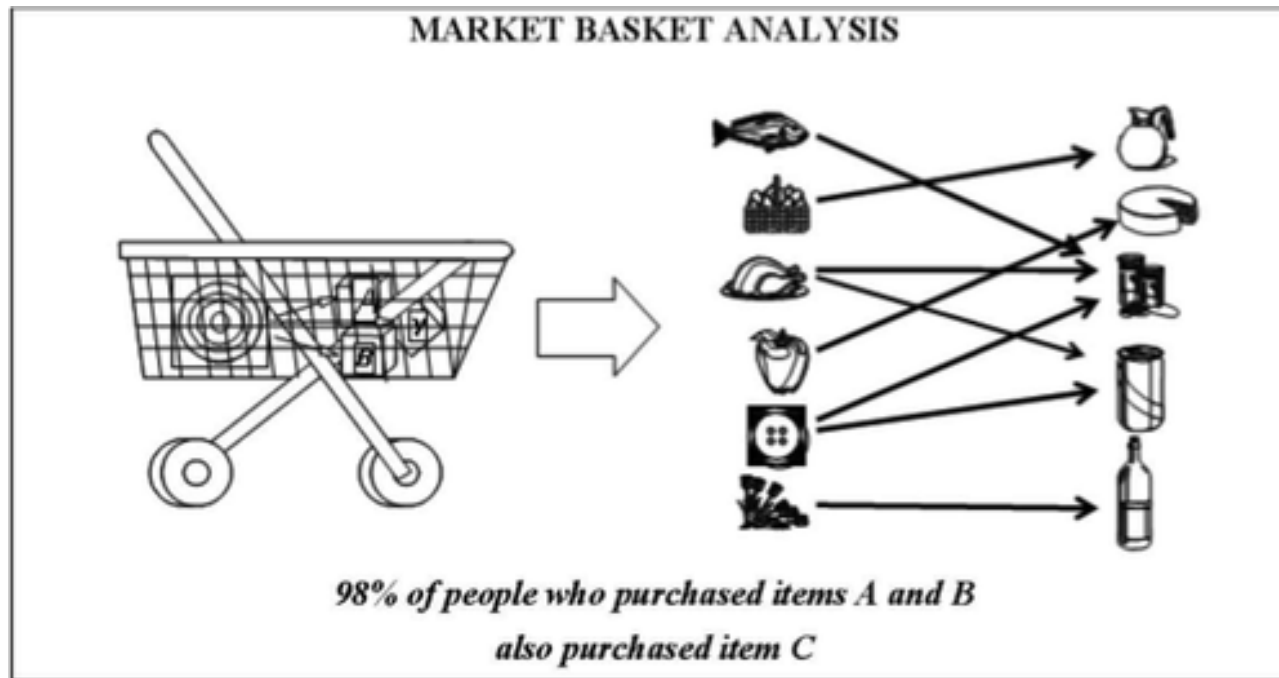
Reading

- Jiawei Han, Micheline Kamber and Jian Pei. (2012) **Data Mining: Concepts and Techniques**, third edition, Morgan Kaufmann. ISBN: 978-0123814791. (or later edition) **Chapter 6**

Outline

- **ARM Background**
- Apriori
- FP-Tree
- Various ARM
- Constraint ARM

Association Rules Mining



Source: <https://au.mathworks.com/>

Association Rules Mining

	I1	I2	I3	I4	I5	I6	I7	I8	I9
ID1									
ID2									
ID3									
ID4									
ID5									

I199999	I200000

ARM - Cartoon



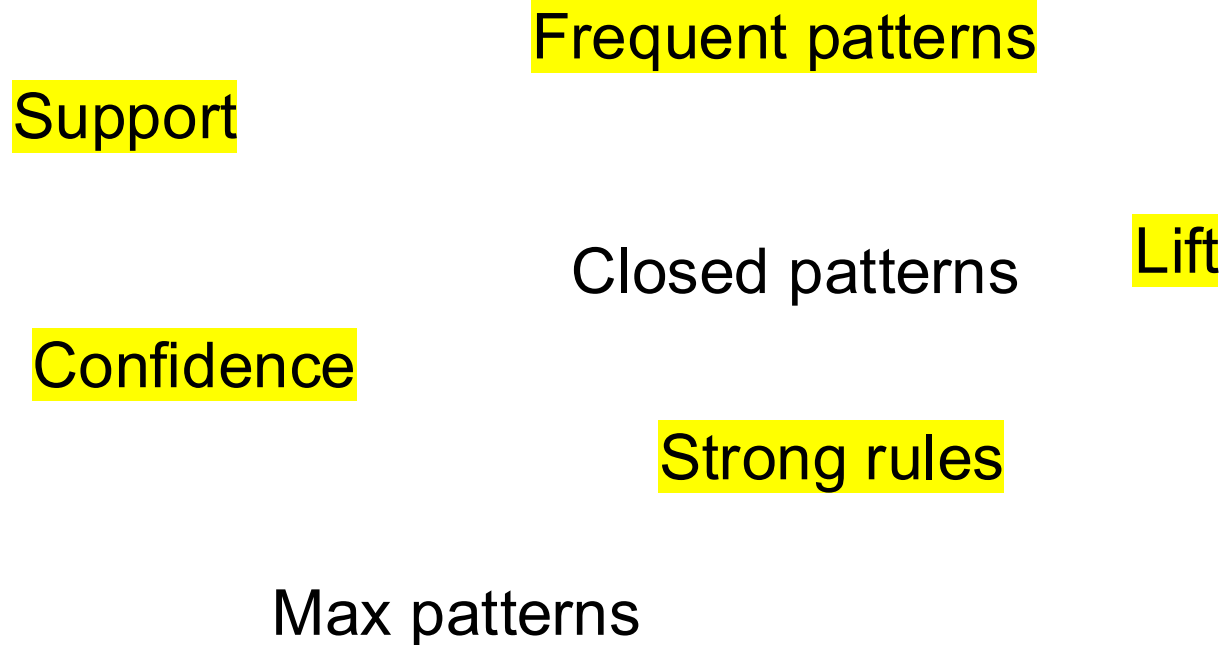
Source: http://gerardnico.com/wiki/data_mining/association/

ARM - Quote

**Choose
your friends
wisely.
Association
is perception.
Perception
is reality.**

Source: <https://au.pinterest.com/pin/150378075037497787/>

ARM – Important Terms



Association Rules Mining

- Concerned with the discovery of significant associations among data items.
- Called as “affinity analysis”, “market basket analysis” or “frequent pattern analysis”
- Finding **inherent regularities** (associations) from frequent itemsets (= association rules)
 - What products were often purchased together?
- Study of “what goes with what”
 - Customers who bought X also bought Y
- Applications
 - Floor/shelf planning, cross-marketing, catalogue design, targeted marketing etc

Used in many recommender systems

Bound Away [Last Train Home](#)



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Transaction Database

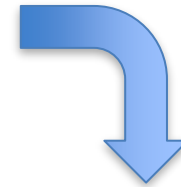
TID	Item
100	Apple
100	Coke
100	DVD
200	Bread
200	Coke
200	Egg
300	Apple
300	Bread
300	Coke
300	Egg
400	Bread
400	Egg

TID (Transaction ID)	Items
100	Apple, Coke, DVD
200	Bread, Coke, Egg
300	Apple, Bread, Coke, Egg
400	Bread, Egg

TID	Apple	Bread	Coke	DVD	EGG
100	1	0	1	1	0
200	0	1	1	0	1
300	1	1	1	0	1
400	0	1	0	0	1

Transaction Database

ID	Income	City	Married	Age	Own House
000	20500	Cairns	Y	21	Y
001	100000	Brisbane	Y	33	N
002	54490	Cairns	N	42	N
...	...	...	...	...	...



TID	Income <50000	Income >50000	Brisbane	Married	Age <40	Age >40	Own House
000	1	0	0	1	1	0	1
001	0	1	1	1	1	0	0
002	0	1	0	0	0	1	0
...	...	...	...	...	...	...	...

Itemsets and Association Rules

- An *itemset* is a set of items
 - E.g., {Apple, Coke, Bread} is an itemset
- A *k-itemset* is an itemset with k items.
- Given a dataset D, an itemset X has a (frequency) *count* in D
- An *association rule* is about relationships between two disjoint itemsets X and Y.

$$X \Rightarrow Y \quad (\text{no overlap between X and Y})$$

X is the *antecedent*, the initial assumption.

Y is the *consequent*, something else that may happen too.

{Apple} $\rightarrow$ {Coke, Bread} is an association rule

- It presents the pattern when X occurs, Y also occurs

Use of Association Rules

- Association rules do not represent any sort of **causality** or **correlation** between the two itemsets.
 - $X \Rightarrow Y$ does not mean X causes Y , so no Causality
 - $X \Rightarrow Y$ can be different from $Y \Rightarrow X$, unlike correlation
- Association rule types
 - **Actionable Rules** – contain high-quality, actionable information
 - **Trivial Rules** – information already well known by those familiar with the business
 - **Inexplicable Rules** – no explanation and do not suggest action

(Trivial and Inexplicable Rules occur most often)

Measuring the strength of a rule

- **Association Rule Mining (ARM)** is finding frequent patterns in a data set.
- A **frequent pattern** (a pattern that occurs frequently in a data set): An intrinsic and important property of a data set.
- Frequent pattern mining can be a foundation for many other essential data mining tasks
- Frequency measuring criteria - **Support** and **Confidence**
- Find all the rules $X \rightarrow Y$ with minimum support and confidence
 - **support**, s , **probability** that a transaction contains X and Y
 - **confidence**, c , **conditional probability** that a transaction having X also contains Y : $P(Y | X) = P(X \cap Y) / P(X)$

Support and Confidence

Person	banana	coffee	milk	chicken
a	1	0	0	0
b	1	1	0	0
c	0	0	0	1
d	1	1	1	1
e	0	1	1	0
f	1	1	1	0

- Association between Banana and Chicken ?
- Association Rule: banana $\rightarrow$ chicken
 - Support =
 - Confidence =

Support and Confidence

Person	banana	coffee	milk	chicken
a	1	0	0	0
b	1	1	0	0
c	0	0	0	1
d	1	1	1	1
e	0	1	1	0
f	1	1	1	0

- Association between Banana and Chicken ?
- Association Rule: banana $\rightarrow$ chicken
 - Support = $1/6$ (Only 1 person got both)
 - Confidence = 25% (Of the 4 banana buyers, 1 also got chicken)

Support and Confidence

Person	banana	coffee	milk	chicken
a	1	0	0	0
b	1	1	0	0
c	0	0	0	1
d	1	1	1	1
e	0	1	1	0
f	1	1	1	0

- Association between Coffee and Milk ?
- Association Rule: Coffee $\rightarrow$ Milk
 - Support =
 - Confidence =

Support and Confidence

Person	banana	coffee	milk	chicken
a	1	0	0	0
b	1	1	0	0
c	0	0	0	1
d	1	1	1	1
e	0	1	1	0
f	1	1	1	0

- Association between Coffee and Milk ?
- Association Rule: Coffee $\rightarrow$ Milk
 - Support = 50% (Half of all people got both)
 - Confidence = 75% (Of the 4 coffee buyers, 3 also got milk)

Support and Confidence

- People want strong associations.
 - Big support $\rightarrow$ the rule includes many people who do both X and Y.
 - High confidence $\rightarrow$ if someone does X, there is a good chance they will also do Y.
- Set limits on what is a “strong” rule.
 1. Minimum support, minimum confidence.
 2. We report all rules that satisfy these limits.

Think/Pair/Share

- What happens with large minimum support?
- What happens with low minimum confidence?
- What are strong rules?
- The question is how to efficiently find all strong rules from a massive transactional DB?

Outline

- ARM Background
- **Apriori**
- FP-Tree
- Various ARM
- Constraint ARM

How to Mine?

- **Downward closure** property
 - Any subset of a frequent itemset must be frequent
 - If {coffee, sugar, banana} is frequent, so is {coffee, sugar} i.e., every transaction having {milk, coffee, sugar} also contains {milk, coffee}
E.g., if {coffee, sugar, banana} occurs 20 times, it would be impossible for {coffee, sugar} to occur < 20 times
- Scalable ARM methods: Two major approaches
 - **Apriori**
 - Frequent pattern growth (FP-growth)

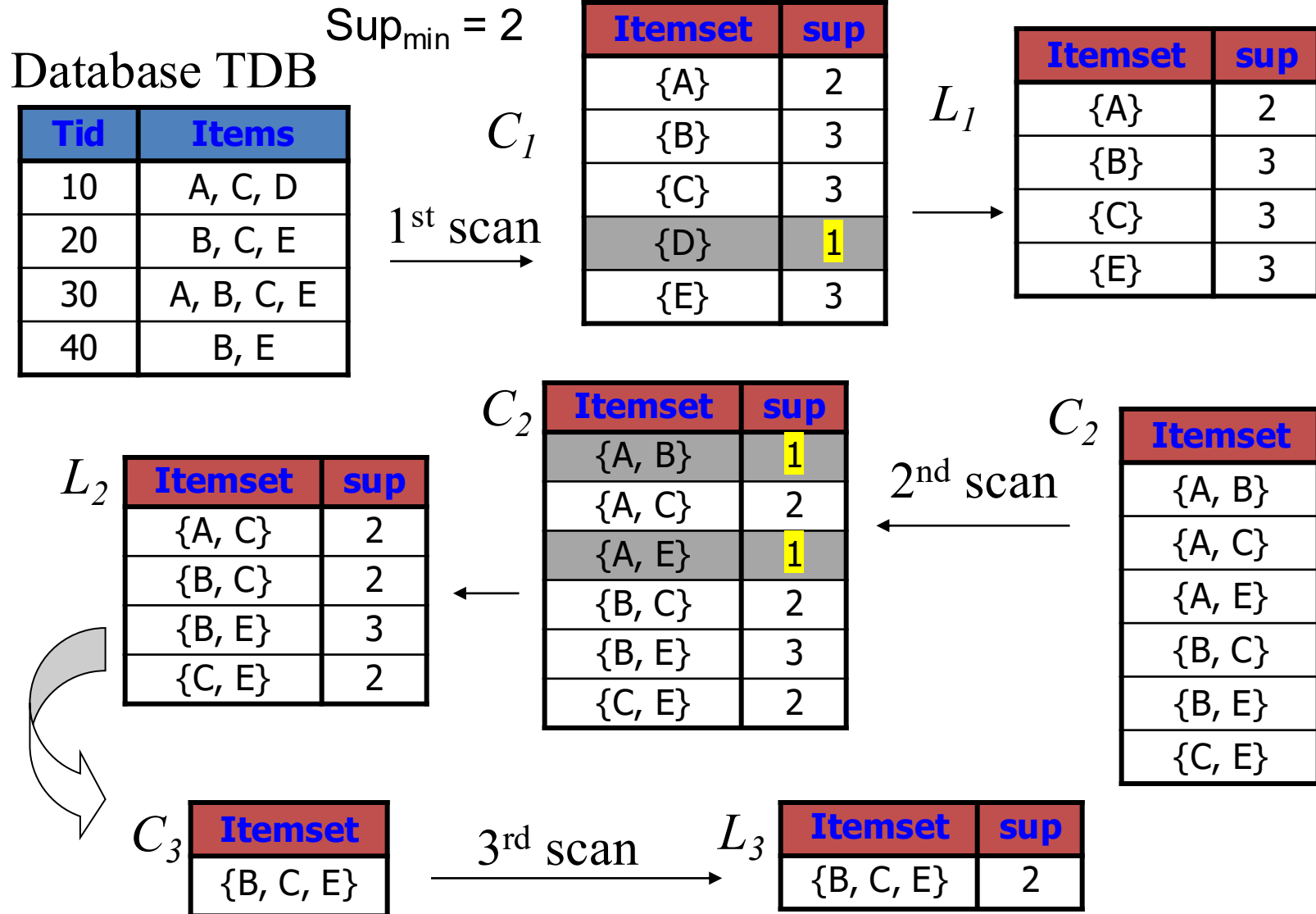
- Apriori pruning principle:

If there is **any** itemset which is infrequent, its superset should not be generated or tested!

- Method:

- Initially, scan DB once to get frequent 1-itemset
- Generate **length (k+1) candidate itemsets** from **length k frequent itemsets**
- **Test the candidates against DB**
- **Terminate when no frequent or candidate set can be generated**

Apriori Algorithm – An Example



Details of Apriori

- Generate candidates
 - Step 1: self-joining L_k
 - Step 2: pruning
- Count supports of candidates
- Example of Candidate-generation
 - $L_3 = \{abc, abd, acd, ace, bcd\}$
 - Self-joining: $L_3 \bowtie L_3$
 - $abcd$ from abc and abd
 - $acde$ from acd and ace
 - Pruning:
 - $acde$ is removed because ade is not in L_3
 - $C_4 = \{abcd\}$

Apriori Rule Generation - Example

Transaction Database

TID	Items
100	Apple, Coke, DVD
200	Bread, Coke, Egg
300	Apple, Bread, Coke, Egg
400	Bread, Egg



Initial Candidates

Itemset	Support
{Apple}	2
{Bread}	3
{Coke}	3
{DVD}	1
{Egg}	3

Apriori Rule Generation - Example

MinSupp:50%
MinConf:90%

Transaction Database

TID	Items
100	Apple, Coke, DVD
200	Bread, Coke, Egg
300	Apple, Bread, Coke, Egg
400	Bread, Egg

C1

Itemset	Support
{Apple}	2
{Bread}	3
{Coke}	3
{DVD}	1
{Egg}	3

F1

Itemset	Support
{Apple}	2
{Bread}	3
{Coke}	3
{Egg}	3

C3

Itemset	Support
{Bread,Coke,Egg}	2

F2

Itemset	Support
{Apple,Coke}	2
{Bread,Coke}	2
{Bread,Egg}	3
{Coke,Egg}	2

C2

Itemset	Support
{Apple,Bread}	1
{Apple,Coke}	2
{Apple,Egg}	1
{Bread,Coke}	2
{Bread,Egg}	3
{Coke,Egg}	2

F3

Itemset	Support
{Bread,Coke,Egg}	2

Apriori Rule Generation - Example

MinSupp:50%

MinConf:90%

Transaction Database

TID	Items
100	Apple, Coke, DVD
200	Bread, Coke, Egg
300	Apple, Bread, Coke, Egg
400	Bread, Egg

C1

Itemset	Support
{Apple}	2
{Bread}	3
{Coke}	3
{DVD}	1
{Egg}	3

F1

Itemset	Support
{Apple}	2
{Bread}	3
{Coke}	3
{Egg}	3

C3

Itemset	Support
{Bread,Coke,Egg}	2

F3

Itemset	Support
{Bread,Coke,Egg}	2

F2

Itemset	Support
{Apple,Coke}	2
{Bread,Coke}	2
{Bread,Egg}	3
{Coke,Egg}	2

C2

Itemset	Support
{Apple,Bread}	1
{Apple,Coke}	2
{Apple,Egg}	1
{Bread,Coke}	2
{Bread,Egg}	3
{Coke,Egg}	2

Frequent Itemsets

Apriori Rule Generation - Example

MinSupp:50%

MinConf:90%

Transaction Database

TID	Items
100	Apple, Coke, DVD
200	Bread, Coke, Egg
300	Apple, Bread, Coke, Egg
400	Bread, Egg



Frequent Itemsets

1-Itemset	2-Itemset	3-Itemset
{Apple}	{Apple,Coke}	{Bread,Coke,Egg}
{Bread}	{Bread,Coke}	
{Coke}	{Bread,Egg}	
{Egg}	{Coke,Egg}	

- All possible Association Rules are:

Apple->Coke

Coke->Apple

Bread->Coke

Coke->Bread

Bread->Egg

Egg->Bread

Coke->Egg

Egg->Coke

Bread->{Coke,Egg}

Coke->{Bread,Egg}

Egg->{Bread,Coke}

{Coke,Egg}->Bread

{Bread,Egg}->Coke

{Bread,Coke}->Egg



Final Association Rules found
(Rules having more than
minimum confidence)

Bottleneck of Apriori

- Challenges
 - Multiple scans of transaction database
 - Huge number of candidates
 - Tedious workload of support counting for candidates
- Improving Apriori: general ideas
 - Reduce passes of transaction database scans
 - Shrink number of candidates
 - Facilitate support counting of candidates

Think/Pair/Share

Imagine, you are a manager of COLES. You do have a lot of potato chips in stock in your warehouse, but you do not have banana in stock.

You are planning to boost the sales of potato chips and also would like to figure out what products would be affected if the store discontinues selling banana.

How ARM could be used for you to answer these business questions?

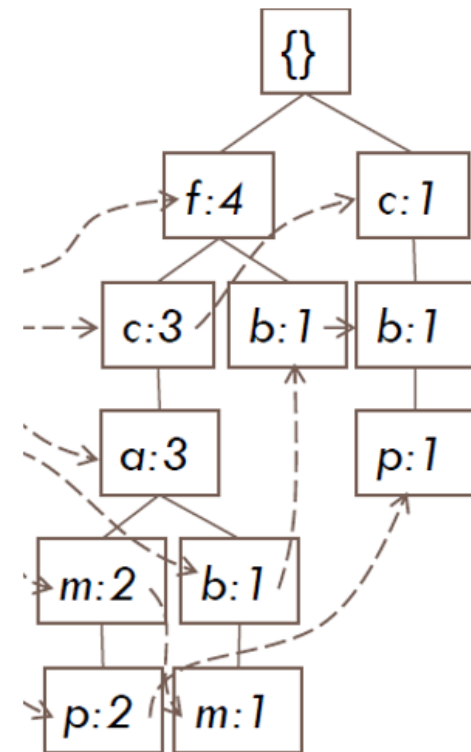
Outline

- ARM Background
- Apriori
- **FP-Tree**
- Various ARM
- Constraint ARM

- Idea: Frequent pattern growth
 - Recursively grow frequent patterns by pattern and database partition
- Method
 - For each frequent item, construct its conditional pattern-base, and then its conditional FP-tree
 - Repeat the process on each newly created conditional FP-tree
 - Until the resulting FP-tree is empty, or it contains only one path—single path will generate all the combinations of its sub-paths, each of which is a frequent pattern

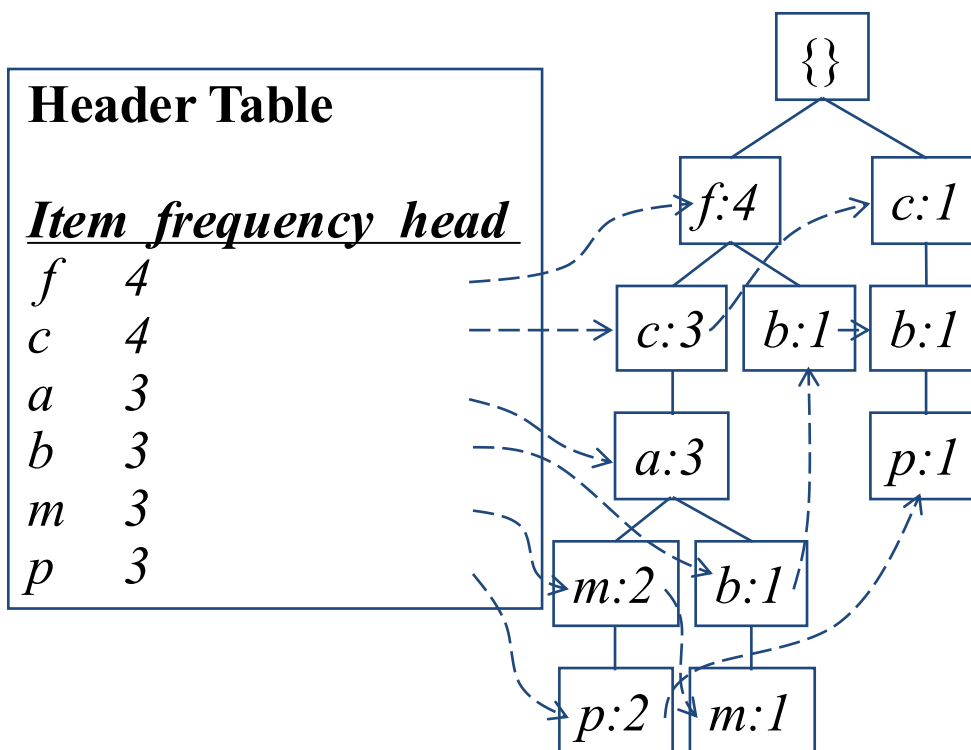
Example for FP-Tree

TID	Items bought
100	{a, c, d, f, g, i, m, p}
200	{a, b, c, f, i, m, o}
300	{b, f, h, j, o}
400	{b, c, k, s, p}
500	{a, c, e, f, l, m, n, p}



FP-Tree

- Starting at the frequent item header table in the FP-tree
- Traverse the FP-tree by following the link of each frequent item p
- Accumulate all of *transformed prefix paths* of item p to form p 's conditional pattern base



Conditional pattern bases

itemcond. pattern base

c $f:3$

a $fc:3$

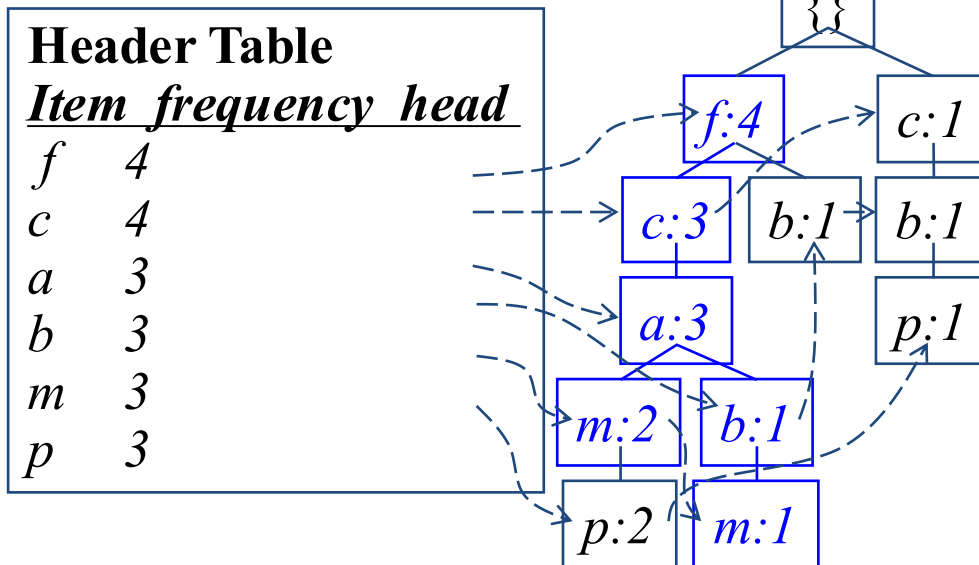
b $fca:1, f:1, c:1$

m $fca:2, fcab:1$

p $fcam:2, cb:1$

FP-Tree

- For each pattern-base
 - Accumulate the count for each item in the base
 - Construct the FP-tree for the frequent items of the pattern base



m-conditional pattern base:
fca:2, fcab:1



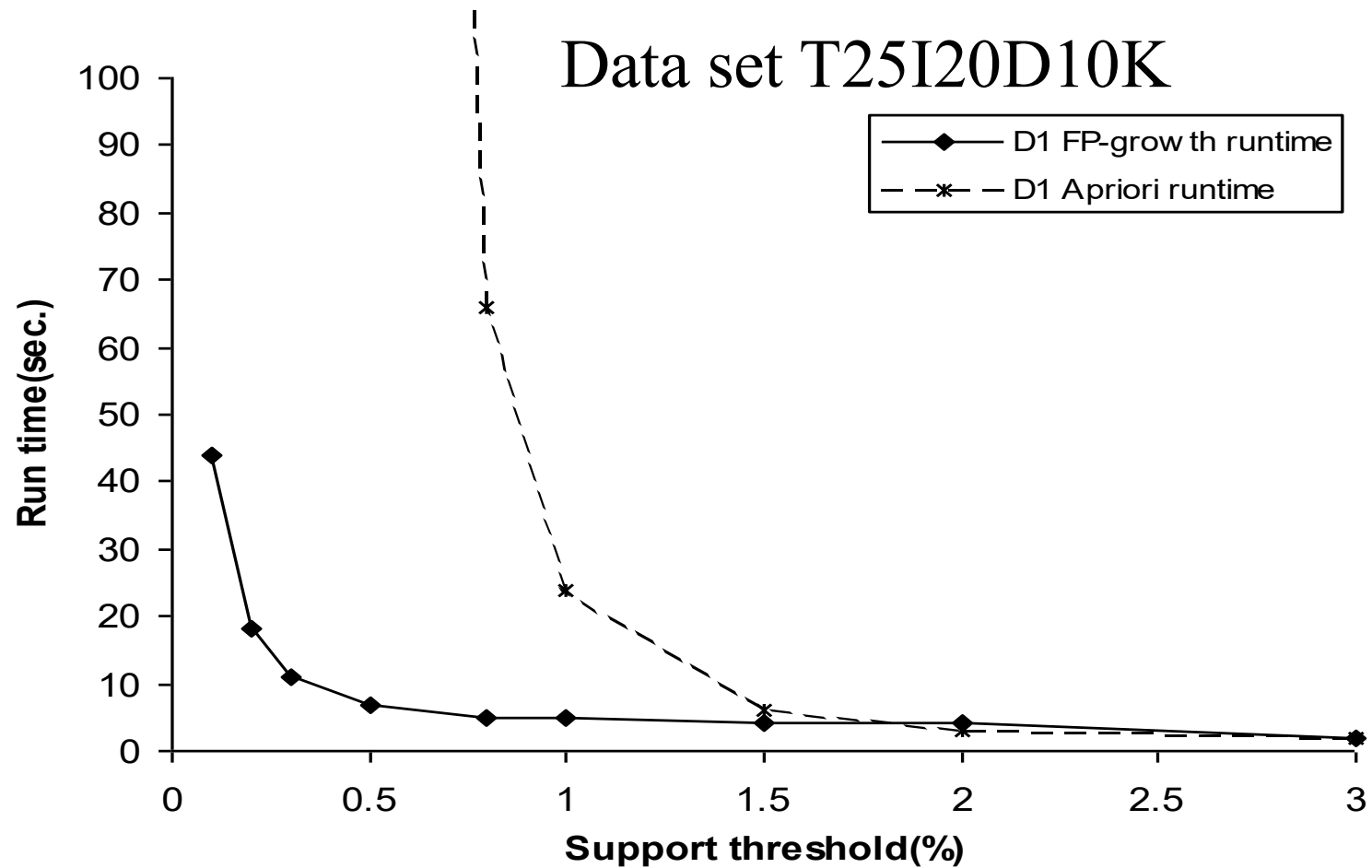
{
|
f:3
|
c:3
|
a:3



All frequent
patterns relate to *m*
m,
fm, cm, am,
fcm, fam, cam,
fcam

m-conditional FP-tree

Apriori vs. FP-tree



Problem: Too Many Patterns

- Any sub-patterns of a frequent pattern is also a frequent pattern.
 - If $\{A, B, C\}$ is a frequent set, $\{A\}$, $\{B\}$, $\{C\}$, $\{A,B\}$, $\{A,C\}$, $\{B,C\}$, $\{A,B,C\}$ are also frequent sets
- A long pattern contains a combinatorially large number of sub-patterns.
- For example, the set $\{a_1, \dots, a_{100}\}$ contains this many possible sub-sets:

$$\binom{100}{1} + \binom{100}{2} + \dots + \binom{100}{100} = 2^{100} - 1 \approx 1.27 \times 10^{30}$$

- Solution: *Mine only **closed patterns** or **max patterns** instead.*
→ This reduces the number of possibilities.

Closed Patterns

- An itemset X is **closed** if:
 - X is *frequent* (has enough support) and
 - There exists *no super-pattern* $Y \supset X$, with the same support as X
- Example of 2 itemsets:
 - X : 50% of people buy milk and coffee.
 - Y : 50% of people buy milk, coffee, sugar.
 - Here, Y contains X , written $Y \supset X$ (similar to $Y > X$) as Y has more items. Y has the same support as X .
 - Thus X is not a closed pattern
- Closed pattern is a lossless compression of frequent patterns (reducing the number of patterns and rules)

Mining Closed Patterns

- Flist: list of all frequent items in support ascending order
 - Flist: d-a-f-e-c
- Divide search space
 - Patterns having d
 - Patterns having d but no a, etc.
- Find frequent closed pattern recursively
 - Every transaction having d also has cfa → cfad is a frequent closed pattern

Min_sup=2

TID	Items
10	a, c, d, e, f
20	a, b, e
30	c, e, f
40	a, c, d, f
50	c, e, f

Max Patterns

- An itemset X is a **max pattern** if X is frequent and there exists no frequent super-pattern $Y \supset X$.
- Mining max-patterns
 - 1st scan: find frequent items
 - A,B,C,D,E
 - 2nd scan: find support for
 - AB, AC, AD, AE: **ABCDE**
 - BC,BD,BE: **BCDE**
 - CD,CE: **CDE**
 - DE:
- Since BCDE is a max-pattern, no need to check BCD, BDE, CDE in later scan

Tid	Items
10	A,B,C,D,E
20	B,C,D,E,
30	A,C,D,F

Potential
max-patterns



Think/Pair/Share

- Exercise. $DB = \{ \langle a_1, \dots, a_{100} \rangle, \langle a_1, \dots, a_{50} \rangle \}$
 - $\text{Min_sup} = 1$.
- What is the set of closed itemset?
 - $\langle a_1, \dots, a_{100} \rangle: 1$
 - $\langle a_1, \dots, a_{50} \rangle: 2$
- What is the set of max-pattern?
 - $\langle a_1, \dots, a_{100} \rangle: 1$
- What is the set of all patterns?
 - !!

Outline

- ARM Background
- Apriori
- FP-Tree
- **Various ARM**
- Constraint ARM

Possible ways of improving performance of the algorithms

- Implementation techniques
 - Use of good data structures
 - Fast implementation of basic operations
- Algorithm improvement
 - Finding algorithms that are more efficient
- Use of parallel processing
- Sampling the transaction databases

Interactive Discovery

In ARM, the user plays an important role in the process

- The user is responsible for setting the initial minimum support and confidence thresholds
- During the discovery, the user may decide to further fine-tune the thresholds
- The user can specify what items are to appear on either or both sides of the resulting rules. (for different purposes)
(e.g.) $\{X\} \rightarrow \{\text{nappies}\}$ or $\{\text{nappies}\} \rightarrow \{X\}$ or $\{\text{Bear}\} \rightarrow \{\text{nappies}\}$
- The user can exploit a category hierarchy of some kind among items.
(e.g.) Instead of “Bread $\rightarrow$ Coke”, “Bakery products” $\rightarrow$ “Soft drinks”

Various Association Rules

- Mining multilevel association
- Mining quantitative association
- Mining interesting correlation patterns

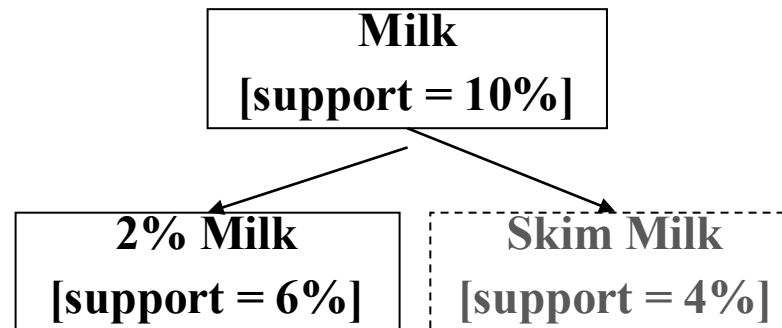
Multi-level ARM

- Items often form hierarchies
- Flexible support settings
 - Items at the lower level are expected to have lower support

uniform support

Level 1
min_sup = 5%

Level 2
min_sup = 5%



reduced support

Level 1
min_sup = 5%

Level 2
min_sup = 3%

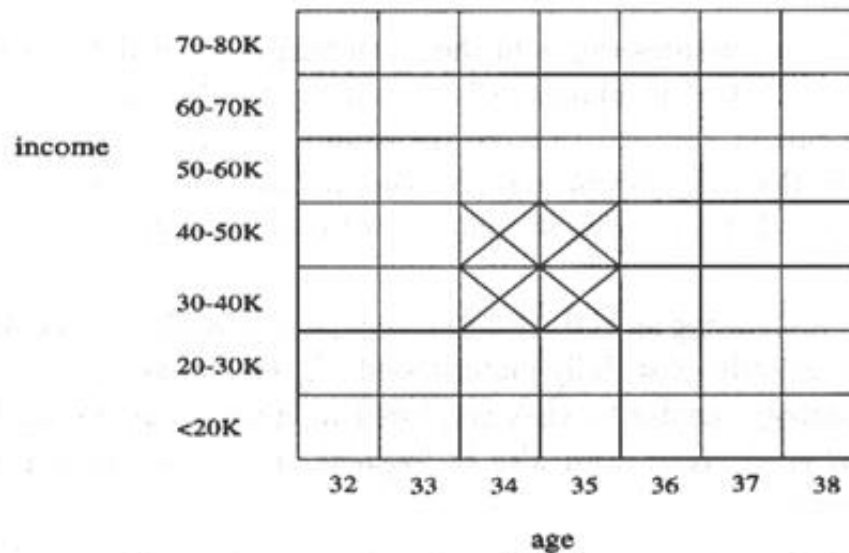
MARM: Redundancy

- Some rules may be redundant due to “ancestor” relationships between items.
- Example
 - milk $\Rightarrow$ wheat bread [support = 8%, confidence = 70%]
 - 2% milk $\Rightarrow$ wheat bread [support = 2%, confidence = 72%]
- We say the first rule is an ancestor of the second rule.
- A rule is redundant if its support is close to the “expected” value, based on the rule’s ancestor.

Quantitative ARM

- Numeric attributes are discretized
- 2-D quantitative association rules: $A_{\text{quan1}} \wedge A_{\text{quan2}} \Rightarrow A_{\text{cat}}$
- Example

$\text{age}(X, "34-35") \wedge \text{income}(X, "30-50K")$
 $\Rightarrow \text{buys}(X, "high\ resolution\ TV")$



Interesting Measure: Lift

- *play basketball* $\Rightarrow$ *eat cereal* [40%, 66.7%] is misleading
 - The overall % of students eating cereal is 75% > 66.7%.
- *play basketball* $\Rightarrow$ *not eat cereal* [20%, 33.3%] is more accurate, although with lower support and confidence
- Measure of dependent/correlated events: [lift](#)

$$lift = \frac{P(A \cap B)}{P(A)P(B)}$$

	Basketball	Not basketball	Sum (row)
Cereal	2000	1750	3750
Not cereal	1000	250	1250
Sum(col.)	3000	2000	5000

$$lift(B, C) = \frac{2000 / 5000}{3000 / 5000 * 3750 / 5000} = 0.89 \quad lift(B, \neg C) = \frac{1000 / 5000}{3000 / 5000 * 1250 / 5000} = 1.33$$

Lift

- If lift is close to 1, this indicates that X and Y are independent of each other, so such rules are **not** interesting
- If $\text{lift} < 1$, X and Y are **negatively correlated**, the presence of X makes Y less likely, even if they occur together frequently
- If $\text{lift} > 1$, X and Y are **positively correlated**, the presence of X makes Y **more likely**

Review

- Given the rule $\{X\} \rightarrow \{Y\}$, how do we calculate
 - Support?
 - Confidence?
 - Lift?
- Does the rule $\{Y\} \rightarrow \{X\}$ have the same confidence as $\{X\} \rightarrow \{Y\}$?

Review

- Given the rule $\{X\} \rightarrow \{Y\}$, how do we calculate
 - Support?
 - $P(X \cap Y)$ (i.e., frequency/dataset size)
 - Confidence?
 - $\frac{P(X \cap Y)}{P(Y)}$
 - Lift?
 - $\frac{P(A \cap B)}{P(A)P(B)}$
- Does the rule $\{Y\} \rightarrow \{X\}$ have the same confidence as $\{X\} \rightarrow \{Y\}$?
 - Not in general

Question

- A dataset contains 10,000 transactions
- Frequencies:
 - {CPU}: 6,000
 - {GPU}: 1,500
 - {CPU, GPU}: 1,200
- For the rule {CPU}→{GPU} determine
 - Support
 - Confidence
 - Lift

Question

- A dataset contains 10,000 transactions
- Frequencies:
 - {CPU}: 6,000
 - {GPU}: 1,500
 - {CPU, GPU}: 1,200
- For the rule {CPU} → {GPU} determine
 - Support
 - $P(\text{CPU and GPU}) = 1200/10000 = 12\%$
 - Confidence
 - $P(\text{CPU and GPU})/P(\text{CPU}) = 1200/6000 = 20\%$
 - Lift
 - $P(\text{CPU and GPU})/P(\text{CPU})P(\text{GPU}) = 0.12/(0.6 \times 0.15) = 1.33$

Think/Pair/Share

Tid	Items
10	A,B,C
20	B,C,D
30	A,C

- Give minimum support = 2 and minimum confidence = 90%
- Find all frequent patterns
- Find all closed patterns
- Find all max patterns
- Find all strong rules

ARM Quiz

Tid	Items
10	A,B,C
20	B,C,D
30	A,C

- Give minimum support = 2 and minimum confidence = 90%
- Find all frequent patterns
- Find all closed patterns
- Find all max patterns
- Find all strong rules

A	2
B	2
C	3
D	1

Frequent 1-itemset

ARM Quiz

Tid	Items
10	A,B,C
20	B,C,D
30	A,C

- Give minimum support = 2 and minimum confidence = 90%
- Find all frequent patterns
- Find all closed patterns
- Find all max patterns
- Find all strong rules

A	2
B	2
C	3
D	1

Frequent 1-itemset

AB	1
AC	2
BC	2

Frequent 2-itemsets

ARM Quiz

Tid	Items
10	A,B,C
20	B,C,D
30	A,C

- Give minimum support = 2 and minimum confidence = 90%
- Find all frequent patterns
- Find all closed patterns
- Find all max patterns
- Find all strong rules

A	2
B	2
C	3
D	1

Frequent 1-itemset

AB	1
AC	2
BC	2

Frequent 2-itemsets

Can ABC be is a possible frequent 3-itemsets?

ARM Quiz

Tid	Items
10	A,B,C
20	B,C,D
30	A,C

A	2
B	2
C	3
D	1

Frequent 1-itemset

AB	1
AC	2
BC	2

Frequent 2-itemsets

- Give minimum support = 2 and minimum confidence = 90%
- Find all frequent patterns
 - A, B, C, AC, BC
- Find all closed patterns
 - C, AC, BC
- Find all max patterns
 - AC, BC
- Find all strong rules
 - $A \Rightarrow C$ (100%), $C \Rightarrow A$ (67%)
 - $B \Rightarrow C$ (100%), $C \Rightarrow B$ (67%)

Outline

- ARM Background
- Apriori
- FP-Tree
- Various ARM
- **Constraint ARM**

- Finding **all** the patterns in a database **autonomously**? — unrealistic!
 - The patterns could be too many but not focused!
- Data mining should be an **interactive** process
 - User directs what to be mined using a **data mining query language** (or a graphical user interface)
- Constraint-based mining
 - User flexibility: provides **constraints** on what to be mined
 - System optimization: explores such constraints for efficient mining—**constraint-based mining**

Constraints in Mining

- Rule constraints can be categorized as
 1. antimonotonic
 2. monotonic
 3. succinct
 4. convertible
 5. inconvertible

Anti-monotonicity

TDB (min_sup=2)

- Anti-monotonicity
 - When an itemset S **violates** the constraint, so does any of its superset
 - $\text{sum}(S.\text{Price}) \leq v$ is **anti-monotone**
 - $\text{sum}(S.\text{Price}) \geq v$ is **not anti-monotone**
- Example. C: $\text{range}(S.\text{profit}) \leq 15$ is **anti-monotone**
 - Itemset ab violates C
 - So does every superset of ab

TID	Transaction
10	a, b, c, d, f
20	b, c, d, f, g, h
30	a, c, d, e, f
40	c, e, f, g

Item	Profit
a	40
b	0
c	20
d	10
e	30
f	30
g	20
h	10

Monotonicity

TDB (min_sup=2)

- Monotonicity
 - When an itemset S **satisfies** the constraint, so does any of its superset
 - $\text{sum}(S.\text{Price}) \geq v$ is **monotone**
 - $\text{min}(S.\text{Price}) \leq v$ is **monotone**
- Example. C: $\text{range}(S.\text{profit}) \geq 15$
 - Itemset ab satisfies C
 - So does every superset of ab

TID	Transaction
10	a, b, c, d, f
20	b, c, d, f, g, h
30	a, c, d, e, f
40	c, e, f, g

Item	Profit
a	40
b	0
c	20
d	10
e	30
f	30
g	20
h	10

- succinct
 - ✓ “All and only those set guaranteed to satisfy the rule can be enumerated”
 - ✓ Itemsets can be generated that satisfy the rule even before the support count starts
 - ✓ Once such subset is generated, iterative testing for the constraint can be effectively avoided
 - ✓ Example : $\min(\text{Item.price}) > 500$ $\max(S) < 120$
 - ✓ $\text{average}(S) > v$, $\text{average}(S) < v$ are not succinct

convertible constraints

- ✓ Constraints not satisfying to any of antimonotonic, monotonic, succinct can be made to satisfy antimonotonic, monotonic constraints by changing order of elements in the set
- ✓ Ex : $\text{Average}(\text{price}) < 100$

Inconvertible

- ✓ Constraints which are not convertible
- ✓ Ex : $\text{Sum}(S) < v$, $\text{Sum}(S) > v$,
element of set S could be any real value

The Apriori Algorithm — Example

Database D

TID	Items
100	1 3 4
200	2 3 5
300	1 2 3 5
400	2 5

Scan D

itemset	sup.
{1}	2
{2}	3
{3}	3
{4}	1
{5}	3

L_1

itemset	sup.
{1}	2
{2}	3
{3}	3
{5}	3

L_2

itemset	sup
{1 3}	2
{2 3}	2
{2 5}	3
{3 5}	2

C_2

itemset	sup
{1 2}	1
{1 3}	2
{1 5}	1
{2 3}	2
{2 5}	3
{3 5}	2

C_2

itemset
{1 2}
{1 3}
{1 5}
{2 3}
{2 5}
{3 5}

C_3

itemset
{2 3 5}

Scan D

itemset	sup
{2 3 5}	2

Naïve Algorithm: Apriori + Constraint

Database D

TID	Items
100	1 3 4
200	2 3 5
300	1 2 3 5
400	2 5

C_1

Scan D →

itemset	sup.
{1}	2
{2}	3
{3}	3
{4}	1
{5}	3

L_1

itemset	sup.
{1}	2
{2}	3
{3}	3
{5}	3

L_2

itemset	sup
{1 3}	2
{2 3}	2
{2 5}	3
{3 5}	2

C_2

itemset	sup
{1 2}	1
{1 3}	2
{1 5}	1
{2 3}	2
{2 5}	3
{3 5}	2

C_2

Scan D ←

itemset	
{1 2}	
{1 3}	
{1 5}	
{2 3}	
{2 5}	
{3 5}	

C_3

itemset
{2 3 5}

Scan D →

L_3

itemset	sup
{2 3 5}	2

Constraint:
Sum{S.price} < 5

The Constrained Apriori Algorithm: Push an Anti-monotone Constraint Deep

Database D

TID	Items
100	1 3 4
200	2 3 5
300	1 2 3 5
400	2 5

Scan D

C_1

itemset	sup.
{1}	2
{2}	3
{3}	3
{4}	1
{5}	3

L_1

itemset	sup.
{1}	2
{2}	3
{3}	3
{5}	3

L_2

itemset	sup
{1 3}	2
{2 3}	2
{2 5}	3
{3 5}	2

C_2

itemset	sup
{1 2}	1
{1 3}	2
{1 5}	1
{2 3}	2
{2 5}	3
{3 5}	2

C_2

itemset	
{1 2}	
{1 3}	
{1 5}	
{2 3}	
{2 5}	
{3 5}	

C_3

itemset
{2 3 5}

Scan D

L_3

itemset	sup
{2 3 5}	2

Constraint:
Sum{S.price} < 5

The Constrained Apriori Algorithm: Push a Succinct Constraint Deep

Database D

TID	Items
100	1 3 4
200	2 3 5
300	1 2 3 5
400	2 5

Scan D

itemset	sup.
{1}	2
{2}	3
{3}	3
{4}	1
{5}	3

L_1

itemset	sup.
{1}	2
{2}	3
{3}	3
{5}	3

L_2

itemset	sup
{1 3}	2
{2 3}	2
{2 5}	3
{3 5}	2

C_2

itemset	sup
{1 2}	1
{1 3}	2
{1 5}	1
{2 3}	2
{2 5}	3
{3 5}	2

C_2

itemset
{1 2}
{1 3}
{1 5}
{2 3}
{2 5}
{3 5}

not immediately to be used

C_3

itemset
{2 3 5}

Scan D

L_3

itemset	sup
{2 3 5}	2

Constraint:
 $\min\{S.\text{price}\} \leq 1$

General Practice

Constraint	Antimonotone	Monotone	Succinct
$v \in S$	no	yes	yes
$S \supseteq V$	no	yes	yes
$S \subseteq V$	yes	no	yes
$\min(S) \leq v$	no	yes	yes
$\min(S) \geq v$	yes	no	yes
$\max(S) \leq v$	yes	no	yes
$\max(S) \geq v$	no	yes	yes
$\text{count}(S) \leq v$	yes	no	weakly
$\text{count}(S) \geq v$	no	yes	weakly
$\text{sum}(S) \leq v \ (a \in S, a \geq 0)$	yes	no	no
$\text{sum}(S) \geq v \ (a \in S, a \geq 0)$	no	yes	no
$\text{range}(S) \leq v$	yes	no	no
$\text{range}(S) \geq v$	no	yes	no
$\text{avg}(S) \theta v, \theta \in \{=, \leq, \geq\}$	convertible	convertible	no
$\text{support}(S) \geq \xi$	yes	no	no
$\text{support}(S) \leq \xi$	no	yes	no

Classification of Constraints

